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Emotion Detection using Speech and Face in Deep Learning

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Abstract - Humans have a unique ability to demonstrate and understand emotions through a variety of models of communication. Based on their emotions or mood swings we can judge whether the human subject is in good psychological condition or not. The most visible apparent deficiencies of today's Emotion capturing systems were their inability to understand the emotions of such patients like mental health disorder, social emotion Agnosia, alexithymia or even autism by using facial expressions. It can be used in schools to help students who find it difficult to express their feelings (introverts) or who have unstable mental health concerns, such as depression, and hence the teacher's or health workers can communicate with their parents and work through their problems. These days, technology allows employers to recognize individuals who are overly stressed in the workplace and release them from their duties. In research work a Deep Learning algorithm is utilized to create an integrated tool to identify the facial emotions and the stress level or emotion quotient from speech. Tools that can assist people in recognizing the emotions of those around them could be very beneficial in treatment settings as well as in regular social encounters. Emotion detection using speech and face in deep learning has made significant progress in recent years, but there are still several challenges that need to be addressed. Here are some of the main challenges: Limited Dataset: The availability of labeled datasets for emotion detection is limited, especially for less common emotions or for specific cultural contexts. This makes it challenging to train deep learning models that can generalize well to new data. Variability in Data: The data used for emotion detection can vary widely in terms of quality, noise, and variability. For example, speech data can be affected by environmental noise, accents, and speaking styles, while facial data can be affected by lighting conditions, facial expressions, and occlusion. Feature Extraction: Extracting relevant features from speech and facial data can be challenging, especially when dealing with complex emotions that are not easily captured by simple features. This requires careful design of feature extraction algorithms and feature engineering techniques. Interpretability: Deep learning models are often seen as "black boxes" that are difficult to interpret. This can make it challenging to understand how the model is making decisions and to diagnose errors or biases in the model.

Ethical and Privacy Concerns: Emotion detection using speech and facial data raises ethical and privacy concerns, as it can be used for sensitive applications such as surveillance, emotion profiling, and behavioral prediction. This requires careful consideration of ethical and privacy issues in the design and deployment of deep learning models for emotion detection.

KEYWORDS: EmotionRecognition ,Alexithymia, S ocial emotional agnosia , Autism

I.INTRODUCTION

Feelings are a crucial part of dayto day lives. It has a fundamental role in all the decisions that we take. It encourages us to coordinateand comprehend the sentiments of others by passingon emotions to other people. In today's time we can easily communicate with computers, laptops, and smart phones through voice operated technologies such as Siri, Google assistant, Alexa, Cortana etc. But the question arises that do these technologies actually know what we are feeling. There is a remarkable advancement in recognizing the speech, but still, we are a little away from building the emotional understanding between a human and a computer. Being able to express and comprehend emotions through a range of communication methods that sets humans apart from other animals. Even the most delicate emotions can be detected by us in any situation, and these feelings constantly inform how we see interpersonal interactions.. Although these algorithms are capable of simple sentiment analysis, they are not equipped to handle the complex emotional content found in everyday speech. In this field, automatic emotion recognition is a natural application.

II.LITERATURE SURVEY

This section discusses the merits and demeritsof the works which are done already related to the proposed work. The author [1] presents a new method for recognizing emotions in conversations using a neural network architecture based on transformers. The research provides an overview of previous work in the field of emotion recognition,

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which has mainly used traditional machine learning techniques with handcrafted features. Recently, deep learning models like RNNs and CNNs have shown promise. However, these models have limitations in modeling conversational dynamics and long-term dependencies. The proposed approach, CTNet, utilizes the transformer model, which has been successful in natural language processing tasks. CTNet takes a conversation's utterance sequence as input and uses self-attention to model the relationships between them. The resulting representation is then fed through a multi-layer perceptron for emotion classification. The authors evaluate CTNet on two datasets: the Friends TV show corpus and the IEMOCAP dataset of acted conversations. They compare the results of CTNet to several baseline models, including RNNs and CNNs, and find that CTNet outperforms them all. Ablation studies are also conducted to analyse the contribution of different components of the architecture. In conclusion, the article presents a novel approach for recognizing emotions in conversations and demonstrates the effectiveness of the CTNet architecture .The research [2] proposes a new method for speech emotion recognition using a neural network architecture combining CNNs, BLSTMs, attention mechanisms, and skip connections .The authors conduct a literature review of previous work in the field of speech emotion recognition and identify the challenges associated with this task. The proposed approach, ASCB-LSTM, achieves state-of-the-art results on two benchmark datasets, outperforming traditional machine learning algorithms and deep learning models without attention and skip connections. Overall, the research presents a promising solution to speech emotion recognition that combines various neural network architectures to improve accuracy in recognizing emotions from speech signals. A novel approach using a transformer-based architecture that integrates multiple modalities for continuous affect recognition [3] has been suggested. The research highlights the challenges in previous work that have focused on recognizing emotions from separate modalities. The proposed MEME-T model achieves state-of-the-art results on two benchmark datasets, outperforming traditional machine learning algorithms and deep learning models without multi-modal attention mechanisms. The results suggest that MEME-T is effective in capturing the dynamic relationships between different modalities and can improve the accuracy of affect recognition. This article[4] presents a novel approach to speech emotion recognition using a multimodal neural network architecture. The authors provide an overview of previous work in the field of speech emotion recognition and identify the limitations of traditional methods that rely solely on audio features. They propose a multimodal approach that combines audio, visual, and textual features using a multi-level multi-head fusion attention-based recurrent neural network (MLMHF-Att-RNN).The MLMHF-Att-RNN model selectively weights the importance of different modalities using attention mechanisms and combines them using a fusion approach that integrates multiple levels and heads. The authors evaluate the model on two benchmark datasets, and the results show that the proposed approach outperforms other state-of-the-art models, achieving significant improvements in accuracy. Overall, the research presents a promising approach to speech emotion recognition that leverages multiple modalities using a novel neural network architecture. The results demonstrate the potential of multimodal approaches in improving the accuracy of speech emotion recognition and highlight the importance of considering multiple modalities in this task. A fusion model that combines textual, audio, and visual features to recognize emotions in multimodaldata [5] has been suggested. The model is based on BERT and is designed to handle input data of different types and from different sources. To evaluate the proposed model, the authors compare its performance with several state-of-the-art emotion recognition models on two benchmark datasets. The results show that the proposed model performs better in terms of accuracy and F1-score.The research's contributions are significant as multimodal emotion recognition is a vital research area with applications in human-computer interaction, affective computing, and mental health monitoring. Overall, the research presents a novel approach that leverages the strengths of BERT and addresses the challenge of fusing heterogeneous features.

The author [6] provides an extensive survey of the field of multimodal machine learning, which involves combining information from multiple modalities, such as vision, speech, and text, to solve complex tasks. The authors present a comprehensive taxonomy of multimodal machine learning approaches, which includes feature fusion, decision fusion, and deep fusion. They also discuss various applications of multimodal machine learning, including emotion recognition, action recognition, and speech recognition. The authors compare and contrast different multimodal machine learning techniques, highlighting their strengths and limitations. They also identify open research challenges in the field, such as handling missing

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modalities, dealing with noisy data, and developing scalable algorithms. Overall, the research provides a valuable resource for researchers interested in multimodal machine learning. The comprehensive survey and taxonomy enable readers to gain a deeper understanding of the field's landscape, identify key challenges, and explore potential solutions. The research's contributions are significant, as multimodal machine learning is an important research area with numerous applications, including human-robot interaction, natural language processing, and multimedia analysis. The research proposes a model for recognizing emotions from speech data, which utilizes Long Short-Term Memory (LSTM) networks with an attention mechanism [7] has been proposed. The model is designed to process short speech segments and identify the underlying emotion. To evaluate the proposed model's performance, the authors conduct experiments on the Emo-DB dataset, which contains recordings of acted emotional speech. The results demonstrate that the proposed model outperforms other state-of-the- art models in terms of accuracy and F1-score.The research's contributions are significant as speech emotion recognition is a crucial research area with a wide range of applications, including affective computing, speech-based human-computer interaction, and mental health monitoring. The proposed model's capability to handle short speech segments and its superior performance make it a promising approach for real-world applications. Overall, the research presents a novel approach to speech emotion recognition that leverages the strengths of LSTM networks and attention mechanisms. The experimental results validate the effectiveness of the proposed model and highlight its potential for practical applications. The research [8] proposes a method for speech emotion recognition that integrates Hidden Markov Models (HMMs) with a deep neural network (DNN) to enhance its performance. To evaluate the proposed method's effectiveness, the authors conduct experiments on the Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset, which contains acted emotional speech recordings. The experimental results show that the proposed methodoutperforms other state-of-the-art models in terms of accuracy, F1-score, and kappa coefficient. The research's contributions are significant as speech emotion recognition is a crucial research area with numerous applications, including affective computing, speech-based human-computer interaction, and mental health monitoring. The proposed method's capability to combine the strengths of traditional machine learning approaches with the power of deep learning makes it a promising approach for real-world applications. Overall, the research presents a novel approach to speech emotion recognition that revisits the traditional HMMs and combines them with a DNN. The experimental results validate the effectiveness of the proposed method and highlight its potential for practical applications. The author proposes a fusion-based approach for recognizing emotions from speech data, which combines both acoustic and text-based features. Theauthors explore several fusion methods [9], including early, late, and hybrid fusion, to evaluate the proposed approach's effectiveness. To evaluate the proposed approach's performance, the authors conduct experiments on two publicly available datasets, the Interactive Emotional Dyadic Motion Capture (IEMOCAP) and the Friends TV Showdataset. The experimental results show that the proposed approach outperforms other state-of-the- art models in terms of accuracy, F1-score, and kappa coefficient. The research's contributions are significant as recognizing emotions from speech is a crucial research area with a wide range of applications, including affective computing, speech-based human-computer interaction, and mental health monitoring. The proposed fusion-based approach's ability to combine both acoustic and text-based features makes it a promising approach for real-world applications. Overall, the research presents a novel approach to recognizing emotions from speech data that utilizes fusion-based methods to combine acoustic and text-based features. The experimental results validate the effectiveness ofthe proposed approach and highlight its potential for practical applications. This author [10] outlines an approach for generalised labelled multi-bernoulli filter-based robust pitch estimation and speaker tracking (GLMB). The proposed method outperforms other state-of-the-art models in terms of accuracy and tracking. The research's contributions are significant for speech processing applications, and the proposed method's potential for practical applications is highlighted.

2.1 EXISTING SYSTEM:

The existing system is Artificial Neural Network – Multilayer Perceptron (MLP). An artificial neural network can be used to learn any nonlinear function. Now, these groups are known as Universal Function Approximates. Learning loads that map any applied equivocally allude to a feed forward artificial neural network. The enactment work was one of the main goals of general guess. Nonlinear features become familiar with the organization through enactment

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capacities. This enables the organization to become adept at any unforeseen relationship between input and output. The Expression MLP is applied equivocally allude to a feed forward artificial neural network.

III.PROPOSED SYSTEM:

As speech is the form of expressing and to connect with the computer world, define speech emotion recognition system. It includes grouping together various methodologies to analyses speech signals so that embedded emotions are identified. There are many models available to process speech signals to make prediction and identify the embedded emotion lying in it. Long Short Term Memory is implemented in this work's recurrent neural network model. Audio file is a sequential data and needed a appropriate model to learn it and analyses. Therefore, select the sequential LSTM model to carry out this work. Use the Haar-cascade algorithm to recognise facial expressions of emotion.

IV.METHODOLOGY:

The goal of emotion detection utilizing speech and face in deep learning methodology is to recognize and categories the emotions that people exhibit through their spoken and facial expressions. In order to analyse and interpret speech patterns and face features, machine learning techniques and neural networks are used.

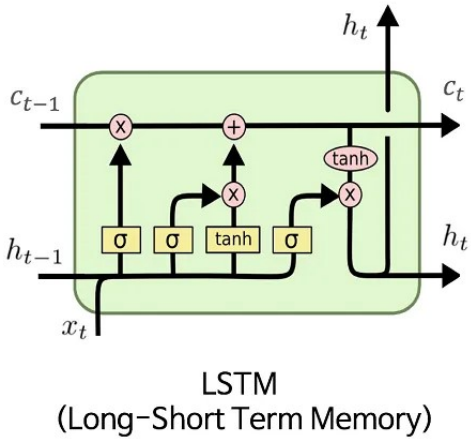

Figure 4.1. LSTM

4.1 SYSTEM ARCHITECTURE:

4.1.1. HOW LTSM MAKES THE PREDICTION?

In order to minimise a loss function, usually the cross-entropy loss, the LSTM model learns to modify the parameters of the gates and the memory cell during training. By feeding new input sequences through the LSTM cells and producing the predicted sequence after the model has been trained, new input sequences can be used to make predictions.

The LSTM model uses the input sequence to generate a hidden state at each timestep by processing it through the LSTM cells. The predicted output is then created by passing this hidden state through a final output layer. A single value, such as a classification label or a numerical value, or a series of values, such as a list of words in a language model, can be the predicted output. Overall, the LSTM model makes predictions based on the input sequence's complex temporal dependencies by using its learned parameters.

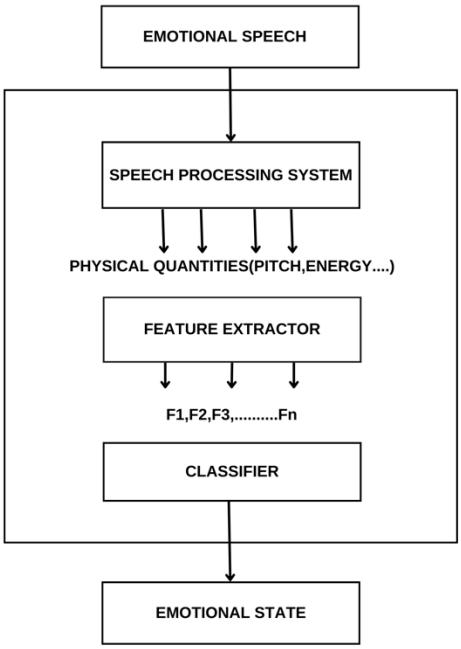

Figure 4.2. Architecture of LTSM

4.1.2. Architecture of LSTM:

A deep learning system called long short-term memory (LSTM) prevents the vanishing gradient problem. features feedback connections that are used to not only create more aesthetically pleasing. In addition to analyzing single data points (such as photographs), it can also analyse entire data sequences (such as audio or video). Back propagated errors were kept from escaping or smashing up thanks to LSTM. I In contrast to typical feed forward neural networks .Instead, LSTM allows faults to propagate backwards across an infinite number of Virtual layers that are spread out in space. Remembering past events that happened hundreds or perhaps millions of distinct time steps ago.. LSTM is usually enhanced with recurrent gates called to as sequential classification was done because here the predictive model is fed with inputs that is distributed over equally spaced time intervals

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and my aim was to forecast the classification group it belongs to. Dense layers were added in the model, dropout layers were also added to reduce over fitting of my prediction model. Keras library supports regularizations of dropout layers. Dropout works by eliminating inputs to a layer, which might be input factors in the information test or enactments from a past layer. It has the impact of recreating countless organizations with totally different organization structure and thus, making hubs in the organization for the most part heartier to the data sources. While adding drop out layers "Relu" activation function was used. A hub's extra weight contribution must be modified by an activation function in order for the hub to act or yield for that input. A piecewise linear function called the "Relu" activation function will give the information directly if it is positive; else, it will give zero. It looks and acts like a linear function,yet is a nonlinear function permitting complex connections in the information to be acquired by themodel. This function utilizes stochastic angle plunge with back propagation of mistakes to prepare profound neural organizations. After the last dense layer "softmax" activation function was used. After that configuration of the model was done for training. All the prepared layers were fed into the LSTM classifier to make the prediction.

4.2 HAAR CASCADE ALGORITHM:

Convolutional and recurrent neural networks, among other deep learning techniques, have recently emerged as the industry standard for recognising emotions. Without the need for manually created features like those used in the Haar Cascade algorithm, these models can learn to extract features from unprocessed image or audio data and classify them directly.

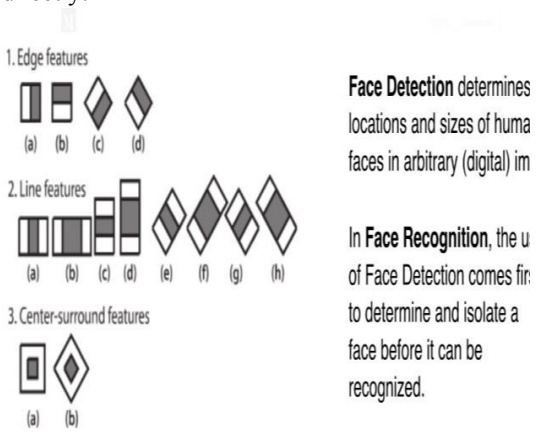

Figure 4.3 Definition of haar cascade algorithm

The Haar Cascade algorithm can nevertheless be helpful as a preprocessing stage for deep learning-based emotion recognition, particularly for face detection and alignment.

To help reduce background noise and make sure that the emotion recognition model is concentrating on the most important part of the image, the Haar Cascade algorithm can be used to identify and crop the face region of an image.

In summary, while the Haar Cascade algorithm is not typically used as the main approach for deep learning-based emotion recognition, it can be a useful preprocessing step for tasks such as face detection and alignment.

A machine learning approach called Haar Cascade is used to find items in an image or video.

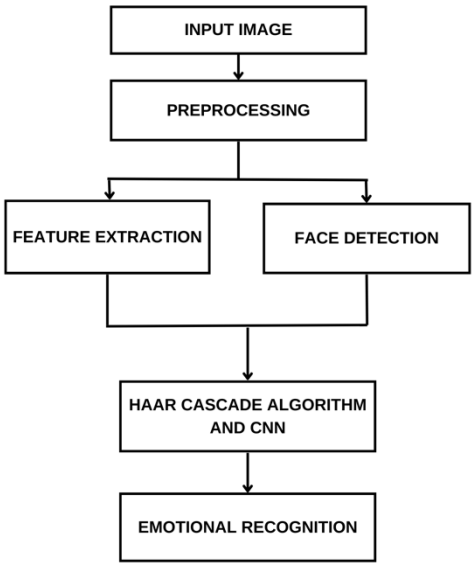

Figure 4.4 System architecture

Use the CNN and Haar-cascade algorithms to identify emotions by analysing facial expressions

4.3 MODULE DESCRIPTION:

4.*3.1 Speech Corpus*

Utilise the "RAVDESS" dataset's Speech sound recordings (16-bit, 48-kHz.wav). 1440 records are included in this portion of the RAVDESS: 24 on-screen characters times 60 preliminary performances for each performer equals 1440. The RAVDESS has 24 knowledgeable on-screen individuals, 12 of whom are male and 12 of whom are female, who each speak two lexically coordinated explanations with a neutral "North

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American" intonation. These were the labels given to the emotions: Neutral, peaceful, joyful, sad, furious, afraid, disgusted, and astonished are the first four letters of the alphabet.

4.3.2 Extraction of features using MFCC

Bringing out the essential attributes is the foremost and the most important step to get the accurate output. It is important to segregate the important and useless attributes from the speech. Important attributes are pitch, tone, etc. Useless features are background noise, disturbances, etc.Human mouth organs like vocal tract, teeth, gums, tongue play an important role in changing the sounds of speech. Different forms of sound are created because of different shapes of these organs. To get an accurate data of a speech it is important to study the impact of the varying shapes of these human organs. Thus, by identifying the frequencies and various other characteristics, can deduce the vocal tract's shape.

4.3.2.1 Steps involved in working of MFCC*:*

Energy increment: This is the first step. In this process the energy of the input voice signal is increased by passing it through a filter bank. Framework: The signals of a sound alter at a very rapid rate when a person is speaking. It is presumed that on a brief time span there is no change in the signals. Therefore, divide the signals into shorter time chunks, such as 20–40 milliseconds. An extremely short time frame is not desirable because more examples would be needed to produce a good result. The signal will vary greatly in the time frame if the time interval is extended, which is also undesirable overlapping: After building up of frameworks, the next task is to overlap the consecutive frameworks. This is done in order to make the signal continuous in nature. Computation: Computation or calculation of power range is the next task. The idea for this mechanism is taken from the working of an ear part that is, cochlea. As a person speaks this organ starts vibrating at different spots. The frequencies approaching sound waves decide these spots. The period-gram gauge it carries out a similar job that is extremely important. It makes out the amount of vitality present around in various frequencies approaching towards it. It is helpful in studying the various frequency regions. This step often makes use of Fast Fourier transformations. Mel scaling: The "Mel scale" is used to get an idea about the building of the gauze banks. These gauzes are of different sizes. The initial one is very narrow that gives the idea about the vitality near to 0 Hertz. As the frequencies are escalated the gauzes get broader to give accurate information about energy athigher frequencies.

Applying Logarithmic function: The next step is to take out their logarithm. In order to understand the audio correctly double the recognized volume of audio. This implies huge variations in energy may not be all that varying if thesound is loud enough.

Discrete Cosine Transformation (DCT): The last step is to process the DCT of the log filter bank energies. The range of its coefficients is between 3-14. There are various other coefficients that are not required. They are disposed off.

4.3.3 Classifier:

Recurrent Neural Network – Long Short Term Memory (LSTM) are used to model tasks where the output is contingent on multiple inputs and dependent on these inputs. Unlike traditional neural networks, RNN operates on a sequence of data inputs and stores the information of the previous inputs. For example, while you are reading a text you don't process individual words, rather you try to remember the previous words and sentences to update the incoming information. In this way you gather a better understanding of the text. RNN works in a similar way. In a speech emotion recognition model, a sequence of voice signals is input in the model and a function is executed at each time step, thus it maintains a state all through.

V.RESULT

The field of emotion recognition is increasingly relying on deep learning algorithms, which have shown promising results. As this technology advances, it is expected that more dependable systems will emerge, opening up countless applications across various industries

In a recent study, researchers investigated several databases and utilized LSTM to tackle the challenge of emotion recognition. The study employed Tensor Flow to train the model, achieving an accuracy rate of 92.88%. With the same architecture, there is potential to develop a real-time emotion recognitionsystem in the future.

Overall, the use of deep learning algorithms in emotion recognition holds great promise for future advancements, and the results of this study demonstrate the potential for accurate and efficient models

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VII.OUTPUT

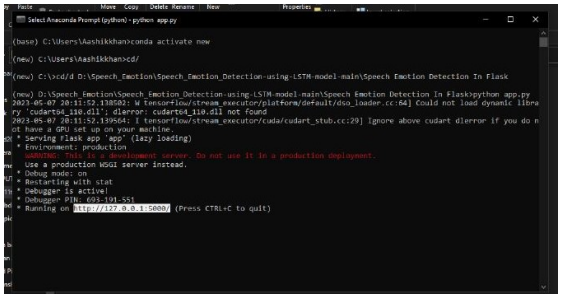

Figure 1 Opening the software using anaconda prompt

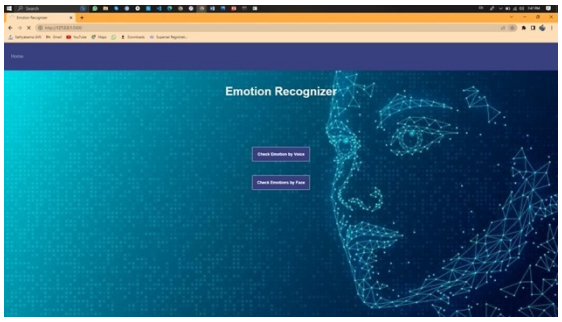

Figure 2 After copying the link provided from the anaconda prompt paste it in a chrome page

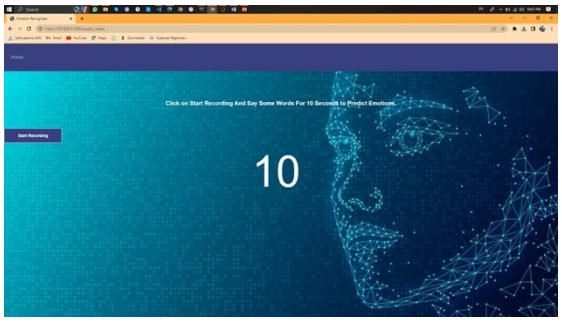

Figure 3 While using input as audio,the timer runs and gets the input

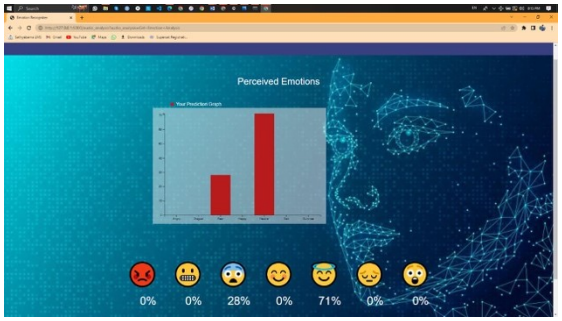

Figure 4 The result of audio input

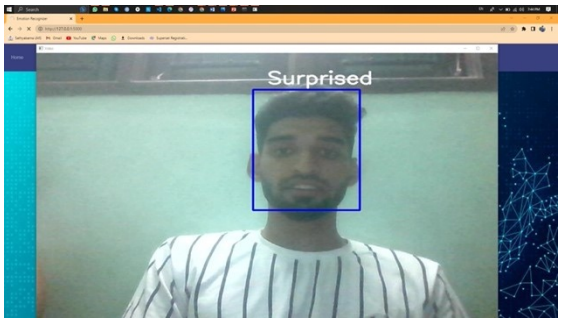

Figure 5 The outcome of a video input

VI.CONCLUSION

Deep learning's use of speech and face to identify emotions has shown encouraging results recently and has the potential to revolutionise a number of applications, including customer service, mental health monitoring, and human-robot interaction. Deep learning models are able to learn to extract pertinent features from speech and facial data and classify emotions with high accuracy, enabling the detection of emotions in real-time and under various circumstances.

Even so, there are still a number of issues to be resolved, such as the scarcity of datasets, data variability, feature extraction, interpretability, and ethical and privacy issues. A multidisciplinary strategy combining developments in deep learning, signal processing, psychology, ethics, and privacy will be necessary to address these problems.

In conclusion, speech and face emotion detection in deep learning has enormous potential for enhancing human-machine interaction and comprehending human emotions. We can anticipate seeing more advanced and precise models that can 7ecognize a wider range of emotions and be used in a growing number of applications as research and development on this topic continues, while also addressing the ethical and privacy issues raised by this technology.

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